

Assignment 6. Midpoint and Trapezoidal method

Marks 20

Posted on 10.09.2026 @ 2:30 pm and due on 10.09.2026 @ 5:40 pm

1. Write library functions for Midpoint and Trapezoidal numerical methods for integration. [10]
2. Using both Midpoint and Trapezoidal method evaluate the following integrals and compare with the analytical results, for $N = 4, 8, 15, 20$. Present the result in tabular form. [10]

$$\int_1^2 \frac{dx}{x} = 0.69314718, \quad \int_0^{\pi/2} x \cos x \, dx = \frac{\pi}{2} - 1, \quad \int_0^1 x \tan^{-1} x \, dx = \frac{\pi}{4} - \frac{1}{2}$$